

FAR WESTERN UNIVERSITY
Semester End Examination-2079

Study Skills in English for Academic Purposes (EAP) SH 125

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Second

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate *full marks*.

Group – A

Attempt any five questions.

(Short Questions)

5×8=40

1. Recognize the differences between pictograms and ideograms. What are their benefits to us?
2. Explain the three basic research methods discussed in the textbook.
3. Explain Hofstede's four cultural dimensions and how they apply to your own culture.
4. Plagiarism is a term used to describe the act of copy. How does it happen, and how can you prevent it?
5. Dissect the discrepancies in gender roles among cultures.
6. Some say that technological communication has displaced face-to-face communication, which is harmful to people. To what extent do you agree? Explain.

Group – B

Attempt any four questions.

(Long Questions)

4×15=60

7. What can a reader do to improve his or her reading efficiency? Consider using examples to illustrate your point.
8. List and briefly describe each of the common frames of assignment structure presented in your textbook.
9. How much do you believe that there is workplace discrimination and that there are fewer women in higher-status occupations than men? Provide evidence to support your point of view or personal experience.
10. The national government is constantly prioritizing. Government funds are limited, and there are numerous conflicting demands for spending. Elucidate.
11. Who are the people recognized as social entrepreneurs? Write about someone you've met or come across in your reading.

FAR WESTERN UNIVERSITY
Semester End Examination-2079
Construction Materials (CE 126)

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Second

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate *full marks*.

Group – A

Attempt any five questions.

(Short Questions)

5×8=40

1. Define the scope of construction materials and their types based on the physical and chemical characteristics. (3+5)
2. Why are joints in concrete structures important? Explain their types. (2+6)
3. What types of diseases you may find in wood? Explain them in brief. What are the properties of lime mortars? (4+4)
4. Suppose your neighbor demands a technical assistance from you regarding selection of a good bricks. At this moment, what must be your role for the selection of good brick and its defects? (8)
5. Why tests are crucial for building stones? Define the procedure of crushing test and water absorption test of stone. (2+3+3)
6. Define high alumina cement and rapid hardening cement. What are the main functions of admixtures? (4+4)

Group – B

Attempt any four questions.

(Long Questions)

4×15=60

7. What are the main conditions favorable for segregation? Define steel and its properties. What are the factors affecting the properties of steel? (5+5+5)
8. Describe the manufacturing process of formation of cast iron. What are the main properties of the compounds of clinker? Describe the paints and their ingredients. (3+7+5)
9. Define the dry process while manufacturing OPC. Describe in brief about roof and floor tiles. Point out the commonly used insulating materials in electrical industry. (4+6+5)
10. What requirements are important for a good location of quarrying stones to be fulfilled? Define curing, its method and effects of improper curing. (5+10)
11. Enumerate the different methods of preservation of timber. Draw a comparison between asphalt, bitumen and tar. (9+6)

FAR WESTERN UNIVERSITY
Semester End Examination-2079
Object Oriented Programming (CT 123)

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Second

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate *full marks*.

Group – A

Attempt any five questions.

(Short Questions)

5×8=40

1. What are the advantages and disadvantages of object oriented programming? Briefly describe the features of C++. (4+4)
2. Define function overloading. Write merits and demerits of inline functions. Write a meaningful program to make a user defined function that illustrates the concept of return by reference. (1+3+4)
3. What is runtime polymorphism? Define abstract class and write syntax for it. How can you show that base class and child class destructors are invoked on deleting base class pointer? Illustrate with suitable example. (1+2+5)
4. What is inheritance? Why it is important in object oriented programming? Write a program to overload unary increment operator (++) as a post fix operator. (1+3+4)
5. What are the advantages of exception handling over conventional error handling? What is function template overloading? Write a suitable program to overload function template with normal function. (4+1+3)
6. Write a program to make a binary file named "student.dat" and write the data (name, roll no, email) of a student into file. Use Random access approach for writing data into the file and reading nth record from the file using concept of object and class. (8)

Group – B

Attempt any four questions.

(Long Questions)

4×15=60

7. A) Briefly explain about the history of C++. Differentiate between procedure oriented programming and object oriented programming. (3+5)

P.T.O.

- B) What are the features of friend function? Write a meaningful program to show that the friend function acts as bridge between two classes to operate on the data of those classes of which the function is friend. (2+5)
8. A) Define Dynamic Memory Allocation (DMA) and write syntax for new and delete operators. Write a program to find the smallest element of an integer array using DMA. (1+2+4)
- B) Write about keywords, identifiers, Namespaces and manipulators. Write a program to perform the arithmetic operations like addition, subtraction, division and multiplication of three integers by making four inline functions for each arithmetic operations using concept of object and class. (4+4)
9. A) What are the constructors and destructors? How the destructors are invoked in reverse order of constructors? Give example. (3+5)
- B) Define objects and classes? Why these are used in C++? Write a meaningful program to copy "engineering" as a content of an object to another object using copy constructor. (2+1+4)
10. A) What is the role of protected access specifier? Explain the hierarchical and multipath inheritance. How constructors and destructors are invoked in multiple inheritance if base and derived class have parameterized constructors. Illustrate with suitable example. (1+3+5)
- B) Define a class named Course and derive three classes as Mathematics, Science and Engineering from the class Course. Then, derive two classes from Science named: Physics and Chemistry. Define data members and member functions as appropriate. Illustrate the concept of member function overriding and accessing overridden member from the derived class in your program. (6)
11. A) What is operator overloading? Explain, how the polymorphism is achieved through operator overloading? What are not overloadable operators in C++? Give example. (1+3+3)
- B) Write a program to make two classes named apple and mango. Use concept of data conversion to convert the apple type data into mango type data (8)

FAR WESTERN UNIVERSITY
Semester End Examination-2079
Engineering Mathematics II (SH121)

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Second

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate *full marks*.

Group – A

Attempt any five questions.

(Short Questions)

5×8=40

8. 1. Define basis and dimension of vector space $V(F)$ with examples. Show that the set of vectors $\{(1, 0, 1), (1, 1, 0), (-1, 0, 1)\}$ forms basis of R^3 . (2+6=8)
9. 2. Prove that the linear equation in x, y, z represents plane surface. Find the distance from the point $(3, 4, 5)$ to the point where the line $\frac{x-3}{1} = \frac{y-4}{2} = \frac{z-5}{2}$ meets the plane $x + y + z = 2$. (4+4=8)
3. With example express your views, why parametric system is assumed to be superior to Cartesian system? Describe the motion of a moving particle $p(x, y)$ whose position at time T is given by $x = \cos t, y = -\sin t, 0 \leq t \leq \pi/2$. (3+5=8)
4. Describe the meaning and application of multiple integral with example. Using the concept of multiple integral find the area of region $R: 0 \leq x \leq 2, x^2 \leq y \leq 2x$. (3+5=8)
5. With suitable example define the linear homogeneous differential equation of first order. Solve the differential equation $(4x + 6y + 5) dy - (3y + 2x + 4) dx = 0$. (2+6=8)
6. Find the complete solution of differential equation $\frac{d^2y}{dx^2} - 4 \frac{dy}{dx} + 4 = x^2 + e^{2x}$. (8)

Group – B

Attempt any four questions.

(Long Questions)

4×15=60

10. 7. Define ordinary point and singular point of differential equation $p_0 \frac{d^2y}{dx^2} + p_1 \frac{dy}{dx} + p_2 y = 0$. Identify the regular singular point of differential equation, $x(x-1) \frac{d^2y}{dx^2} + (3x-1) \frac{dy}{dx} + y = 0$. Also find power series solution of the differential equation about $x=0$. (3+2+10=15)

P.T.O.

8. Define cone and cylinder. Deduce the equation of cone with vertex at (α, β, γ) and guiding curve is $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0, z = 0$. Also find the cone with vertex at $(3, 2, 1)$ and base is $2x^2 + 3y^2 = 1, z = 1$. (2+8+5=15)

9. (a) Define homogeneous and non-homogeneous system of linear equation with example. Test the consistency of system of linear equations, $x + y + z = 4, x + y + 2z = 2, 2x + 2y + z = 5$. (2+6=8)

- (b) Show that the linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (x, x+y)$ is invertible and also find formula for T^{-1} . (5+2=7)

10. (a) Evaluate the integral $\int_0^a \int_0^{\sqrt{a^2-x^2}} \sqrt{x^2 + y^2} \, dy \, dx$, by changing into polar co-ordinate. (8)

- (b) Describe the meaning of $\int_{\theta=\alpha}^{\theta=\beta} f(\theta) d\theta$. Using the same concept find the area of region that lies inside the circle $r = 1$ and outside cardioid $r = 1 - \cos\theta$. (2+5=7)

11. (a) Prove that the plane $2x - y + 2z - 14 = 0$, touches the sphere, $x^2 + y^2 + z^2 - 4x + 2y - 4 = 0$. Also find the point of contact. (5+2=7)

- (b) Define Eigen value and Eigen vector with examples. Find the Eigen values and Eigen vectors of the matrix $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$. (2+6=8)

FAR WESTERN UNIVERSITY
Semester End Examination-2079
Physics (Mechanics and Optics) SH 122

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Second

Full Marks: 100

Time: 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate *full marks*.

Group – A

Attempt any five questions.

(Short Questions)

5×8=40

1. State Bernoulli's theorem. Show how Bernoulli's theorem is applied to measure the rate of discharge through the city water main's.
2. What are the optical devices require producing circularly polarized light from unpolarized light? Explain how they are used to produce circularly polarized light.
3. Define interference of light and coherent sources. Explain the techniques for creating coherent sources of light.
4. Describe the working of Ruby laser.
5. Define resolving power. Derive the expression for resolving power of a diffraction grating.
6. Compare the loads required to produce equal depressions for two beams made of same material and having the same length and weight with only difference that one has circular cross-section while the cross-section of the other is square.

Group – B

Attempt any four questions.

(Long Questions)

4×15=60

7. (a) Define Young's modulus (Y), Bulk modulus (K) and Poisson's ratio (σ). Derive the relation between them. (10)
(b) What couple must be applied to a wire 1 m long, 1mm in diameter in order to twist one end of it, through 90° , the other end remaining fixed. Modulus of rigidity of material of the wire is $2.8 \times 10^{10} \text{ N/m}^2$. (5)
8. (a) Define surface tension and angle of contact. Determine the expression for excess pressure inside a drop. (10)
(b) A water drop is observed to fall through a gas of density 0.001 gm/cc with a constant velocity of 98 cm/s . What is the radius of the drop? The coefficient of viscosity of the gas is $2 \times 10^{-4} \text{ poise}$. (5)

9. (a) Describe Michelson interferometer with diagram. Explain how the wavelength of light can be determined with its help. (10)
- (b) Plane polarized light passes through a calcite plate with its optic axis parallel to the faces. Calculate the least thickness of the plate for which the emergent beam will be plane polarized. Given $\mu_o = 1.6584$ and $\mu_e = 1.486$ and wavelength of light is $5000 \text{ \AA}$ (5)
10. (a) Define neutral surface and bending moment of a beam. Deduce the relation for bending moment of a beam. (10)
- (b) Capillary tube of length l , $2l$ and $l/2$ are connected in series, the radii being r , $r/2$ and $r/3$ respectively. If a streamline flow is maintained and the pressure across first capillary is P_1 , deduce pressure across the second and third capillaries. (5)
11. (a) Differentiate between Fresnel and Fraunhofer diffraction. Explain Fraunhofer diffraction of light through single slit. (10)
- (b) Calculate the numerical aperture and acceptance angle of an optical fibre from the following data: $\mu_1(\text{core}) = 1.55$ and $\mu_2(\text{clad}) = 1.50$ (5)

FAR WESTERN UNIVERSITY
Semester End Examination-2079

Fundamentals of Thermodynamics and Heat Transfer (ME127)

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Second

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate *full marks*.

Full Marks: 100

Time: 3hrs.

440 KPa

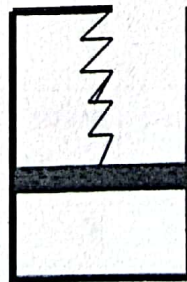
Group – A

Attempt any five questions.

(Short Questions)

5×8=40

1. Define macroscopic and microscopic viewpoint as applied to the study of thermodynamics. Also list their features. Differentiate between heat transfer and work transfer. Derive the mathematical expression for transfer for an isobaric process. (4+4)
2. Define thermodynamic state and thermodynamic process. A piston-cylinder device as shown in figure contains 0.05m^3 of a gas initially at 200kPa . At this state, a linear spring that has a spring constant of 150kN/m is touching the piston but exerting no force on it. Now heat is transferred to the gas causing the piston to rise and to compress the spring until the volume inside the cylinder triples. If the cross-sectional area of the piston is 0.25m^2 , determine the final pressure inside the cylinder. (2+6)



3. Define pure substance. Explain why quality is necessary to define the state of two phase mixture and derive an expression for specific volume of a two phase mixture in terms of quality. (2+3+3)

4. A power-plant operating on an ideal Brayton cycle delivers a power output of 80MW. The minimum and maximum temperatures during cycle are 300K and 1500K respectively. The pressure at the inlet and exit are 100kPa and 1400kPa respectively.
- Determine the thermal efficiency of the cycle
 - Determine the power output from the turbine and
 - What fraction of the turbine power output is required to drive the compressor? [Take $C_p = 1005 \text{ J/kg K}$, $\gamma = 1.4$] (8)
5. Derive the general mass conservation and energy conservation equations for control volume. (3+5)
6. A steel pipe having an outside diameter of 2 cm is to be covered with two layers of insulation, each having a thickness of 1 cm. The average conductivity of one material is 5 times that of the other. Assuming that the inner and outer surface temperatures of the composite insulation are fixed, calculate by what percentage of heat transfer will be reduced when the better insulating material is next to the pipe than it is away from the pipe. (8)

Group – B

Attempt any four questions. (Long Questions) 4×15=60

7. Define entropy and isentropic process. Derive the mathematical expression for entropy relation for an ideal gas in terms of pressure and temperature and temperature and volume. Fill in the blanks in the following table with the corresponding properties of water. (2+5+8)

State	P, kPa	T, °C	x, %	v, m ³ /kg	h, kJ/kg	Degree of Superheat
1	300	133	82	0.5	2346	
2	476	150	60	0.23	1900	
3	200	200	0	0.0011	855	
4	1000	179	60	0.11	2000	

8. Define power cycle and refrigeration cycle. Sketch the Rankine cycle on P-v and T-s diagrams and derive the expression for efficiency. A heat Pump having COP of 5 maintains a building at a temperature of 24°C by supplying heat at a rate of 72000kJ/h , when the surroundings is at 0°C . The heat Pump runs 12 hours a day and electricity costs Rs 10/KWh.
- Determine the actual and minimum theoretical cost per day.
 - Compare the actual operating cost with the cost of direct electric resistance heating. (2+6+7)
9. Define the three modes of heat transfer. Derive an overall heat transfer coefficient for a hollow cylinder subjected to convection medium on both sides. The compression ratio of an air standard Otto cycle is 8. At the beginning of the compression process, the pressure and temperature of air are 100kPa and 20°C respectively. The heat added per kg of air during the cycle is 2000kJ/kg . Determine:
- the pressure and temperature at the end of each process of the cycle.
 - the thermal efficiency and
 - the mean effective pressure. [Take $\gamma = 1.4$, $C_v = 718\text{J/kg K}$] (3+5+7)
10. State and explain Kelvin-Planck and Clausius statement of second law of thermodynamics. Also prove their equivalence. Air expands through an adiabatic turbine from 1000kPa and 1000K to 100kPa and 400K . The inlet velocity is 10m/s whereas exit velocity is 100m/s . The power output of the turbine is 3600kW . Determine the mass flow rate of air and the inlet and exit areas. [Take $R = 287\text{J/kgK}$ and $C_p = 1005\text{J/kgK}$]. (4+4+7)
11. Explain the working principle of an ideal Diesel cycle. Sketch the cycle on P-v and T-s diagrams and derive expression for its efficiency in terms of compression ratio and cut-off ratio. A rigid vessel having a volume of 0.4m^3 contains 2.0kg of liquid water and water vapor mixture in equilibrium at a pressure of 250kPa . Calculate:
- The volume and mass of liquid.
 - The volume and mass of vapor
 - Temperature
 - Enthalpy
 - If it is heated until its pressure reached to 350kPa , what will be its quality? (8+7)

FAR WESTERN UNIVERSITY
Semester End Examination-2079
Engineering Drawing II (AR124)

Faculty: Engineering (Civil)

Level: Undergraduate

Semester: Second

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate *full marks*.

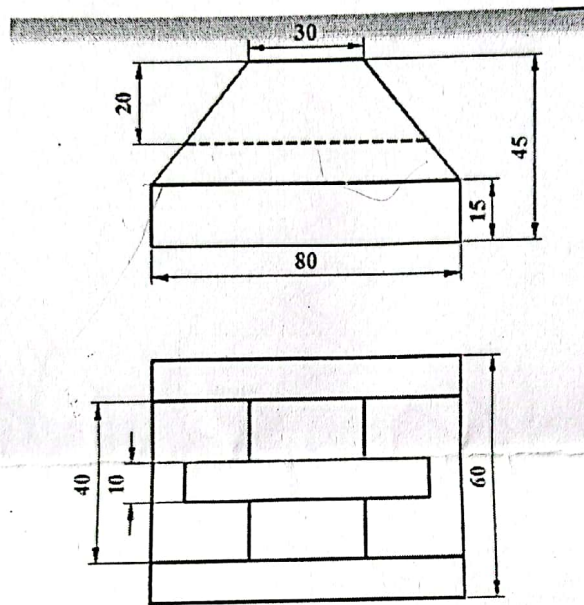
Assume suitable data if necessary.

Full Marks: 60

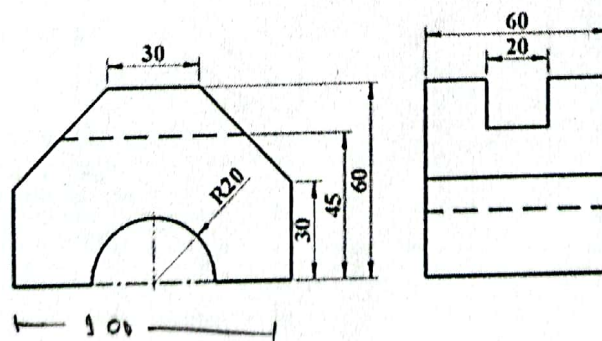
Time: 3hrs.

Group – A
Attempt all questions.

1. Draw the isometric drawing of the following orthographic projections given in the figure. [6]



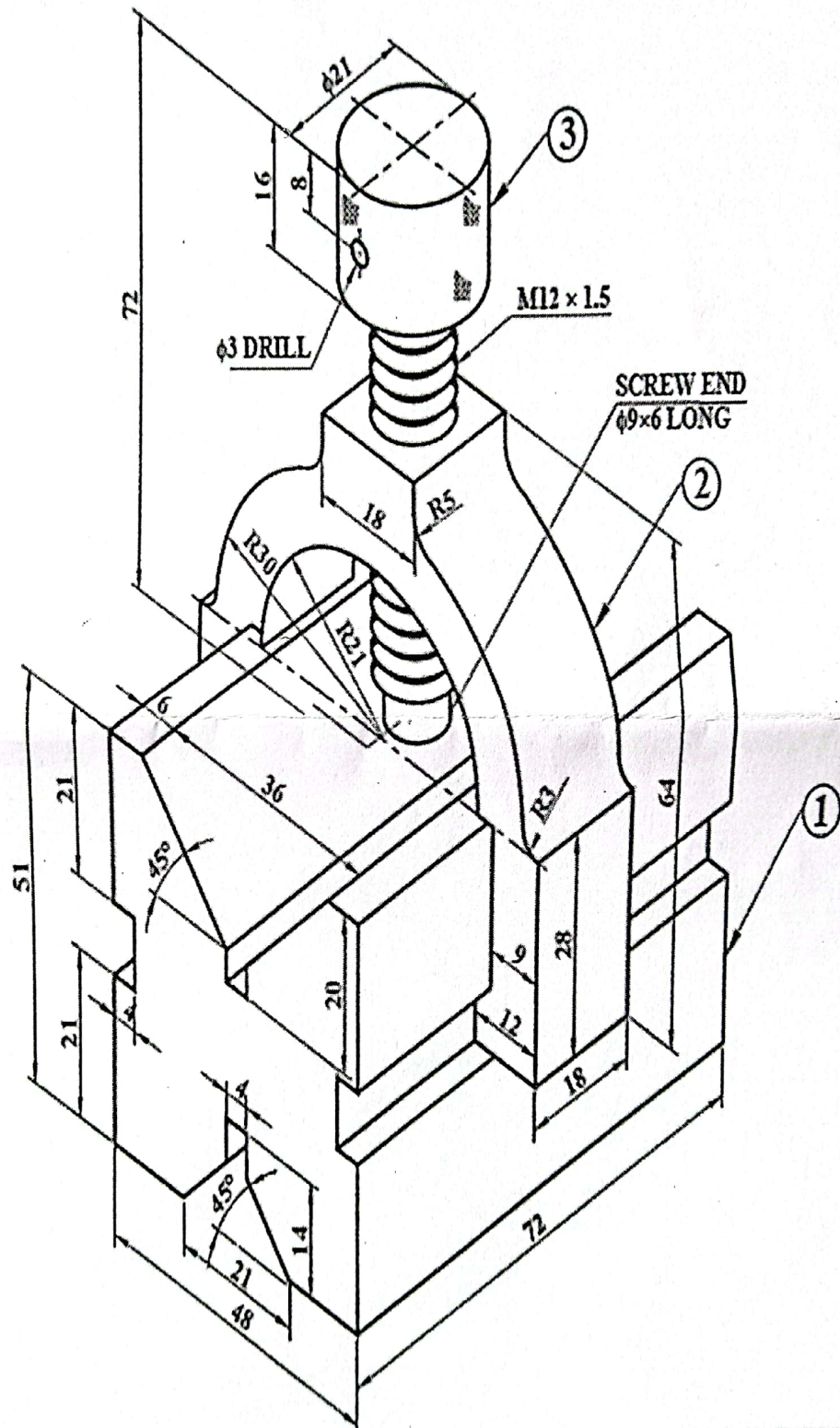
2. Draw the cavalier oblique drawing of the given orthographic projections in the figure. [8]



3. Draw all the views of HEX M 24×3×40/60 bolt. [8]
4. Determine the minimum and maximum sizes, allowance, type of fit and system of given hole-shaft assembly 60 S6/h12 along with a neat diagram. [Given: $S = -0.053$, $6 = 0.019$, $h = 0.000$, $12 = 0.300$] [8]

5. Make a complete set of detail drawings for a V-block Clamp shown in Figure.

[15]



6. Draw the front view with section of the assembly with the detail drawing shown below. [15]

